

Degrees-of-freedom proof (Grubler's equation)

Use Grubler's equation to calculate DOF calculation

$DOF = 3(L - 1) - 2J$, L = total number of links, J = total number of revolute (pin) joints

Build the mechanism up piece by piece, and check that each piece you add doesn't change the DOF.

Counting the links (L):

The frame is link 1. The crank is link 2. Coupler J (crank pin to rocker 1) is link 3. Rocker 1 is link 4. Coupler K (crank pin to rocker 2) is link 5. Rocker 2 is link 6. The dyad link running from point D to the new foot joint is link 7. The dyad link running from point C through to the foot tip is link 8.

That's $L = 8$ total links, so $L - 1 = 7$ moving links.

Counting the joints (J):

Start at O2: frame meets crank, that's 1 joint. At the crank pin, three links meet (crank, coupler J, coupler K) — three links at one point counts as 2 joints, not 1, so add 2. Where coupler J ends, it meets rocker 1: 1 joint. At O4, frame meets both rocker 1 and rocker 2 — again three links at one point, so that's 2 joints. Where coupler K ends, it meets rocker 2: 1 joint. At point D, rocker 1 meets the first dyad link: 1 joint. At point C, rocker 2 meets the second dyad link: 1 joint. And where the two dyad links meet each other to form the new joint carrying the foot: 1 joint.

Adding those: $1 + 2 + 1 + 2 + 1 + 1 + 1 + 1 = J = 10$.

Plugging into the formula:

$$DOF = 3(8 - 1) - 2(10)$$

$$DOF = 3(7) - 20$$

$$DOF = 21 - 20$$

$$DOF = 1$$

This formula proves that DOF is one and we use one actuator.